🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 5 · STANDARD 6.GR.4

Represent Three-Dimensional Figures in Two Dimensions

Lesson 5-6 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes

Level 2 Standard

TODAY'S OBJECTIVES

Content Objective: I can find the surface area of a solid by using its net.
Language Objective: I can explain my work using the words surface area, net, face, and two-dimensional.

See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

Fill in each blank as we go. Use the Word Bank to help you.

WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Surface area | Net | Face | Two-dimensional | Edge

Tap any word to see its meaning and a picture.

1. The total area of all the faces of a solid is its ________.

2. A flat pattern that folds up into a solid is a ________.

3. A flat surface of a solid figure is a ________.

4. A flat shape with length and width but no thickness is ________.

5. A line segment where two faces meet is an ________.

Write About the Math

How many sheets of wrapping paper are needed to cover the entire box?
¿Cuántas hojas de papel de regalo se necesitan para cubrir toda la caja?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Surface area** — *Área de superficie*
- ☐ **Net** — *Plantilla (desarrollo plano)*
- ☐ **Face** — *Cara*
- ☐ **Two-dimensional** — *Bidimensional*
- ☐ **Edge** — *Arista*

Model: *A net lays every face flat so you can see and measure all six faces.* — Students explain a rectangular prism has 6 faces, and the net unfolds the solid into a flat (two-dimensional) pattern so every face is visible to measure.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The box's surface area is ____ in² because SA = 2(____×____) + 2(____×____) + 2(____×____) = ____ + ____ + ____ = ____. You need ____ sheets because ____ ÷ 400 ≈ ____.**
- **My answer is ____ because ____.**

  Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**

  Mi afirmación es ____.
- **My evidence is ____, so ____.**

  Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Surface area
2. **Notes 2:** Net
3. **Notes 3:** Face
4. **Notes 4:** Two-dimensional
5. **Notes 5:** Edge
6. **Watch for this mistake:** *A common mistake in Surface Area Using Nets is finding the area of only one face from each of the 3 matching pairs and forgetting to double it, instead of counting all 6 faces. For a box that is 9 in × 4 in × 5 in, a student might add just one face from each pair: (9×4) + (9×5) + (4×5) = 36 + 45 + 20 = 101 in², but the net actually has two of each face, so the correct total is 2(36) + 2(45) + 2(20) = 72 + 90 + 40 = 202 in² — exactly double. Before you submit, ask: 'Did I count both matching faces in each pair, or only one from each pair?'*
7. **Try It 1:** D. 52 in² — *From the net: SA = 2(2×3) + 2(2×4) + 2(3×4) = 12 + 16 + 24 = 52 in².*
8. **Try It 2:** — *When a net folds up, each flat shape becomes one face of the solid, so the number and the shapes of the faces identify the solid — six identical squares fold into a cube, while three pairs of matching rectangles fold into a rectangular prism.*